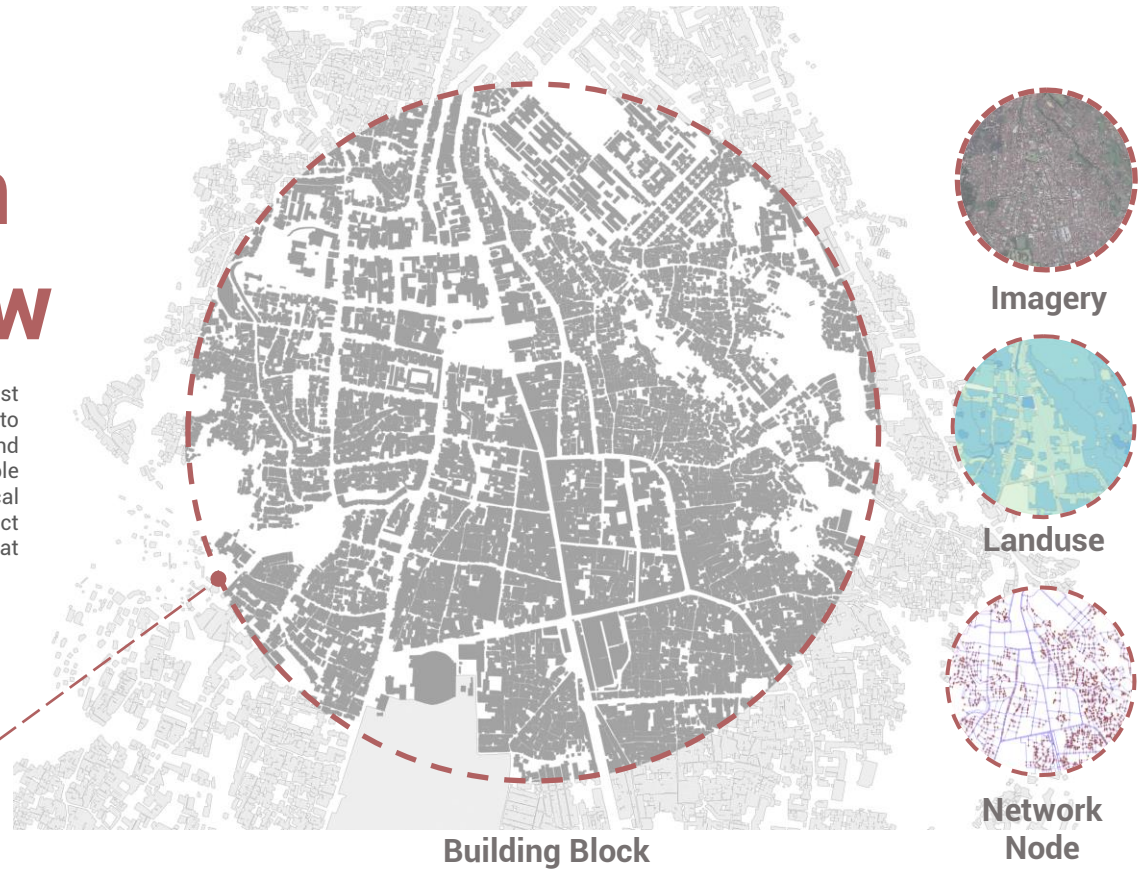


Impact of City Morphology on Pedestrian Flow

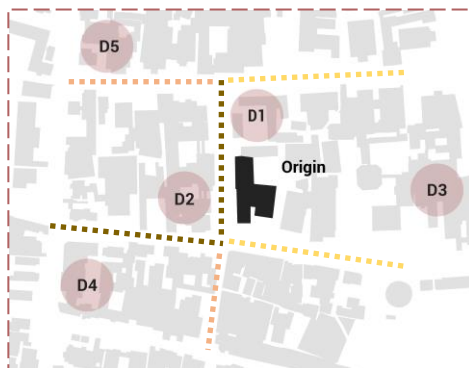
City governments and urban planners frequently aim to boost pedestrian engagement along city streets, integrating this into larger sustainability initiatives, community cohesion, and economic growth plans. Though walkability has received ample attention in planning literature, most planners still lack practical methods for predicting how development proposals could affect pedestrian activity on specific streets or public spaces at different times of the day.

Study Area Magelang City



Methodology uses is Betweenness Algorithm

How many times each link is crossed if trips are made along the shortest paths from a set of origins to a set of destinations.



Capture trips between separate origins and destinations. This allows betweenness results to be estimated from. This analysis also used weighted factor of landuse.



A search radius variable is implemented, which enables to determine the maximum network distance between origins and destinations for which trips are modeled. A search radius constraint allows trips from each place.

Theory Review

Degree

Number of connection?
"A" has a high degree

Closeness

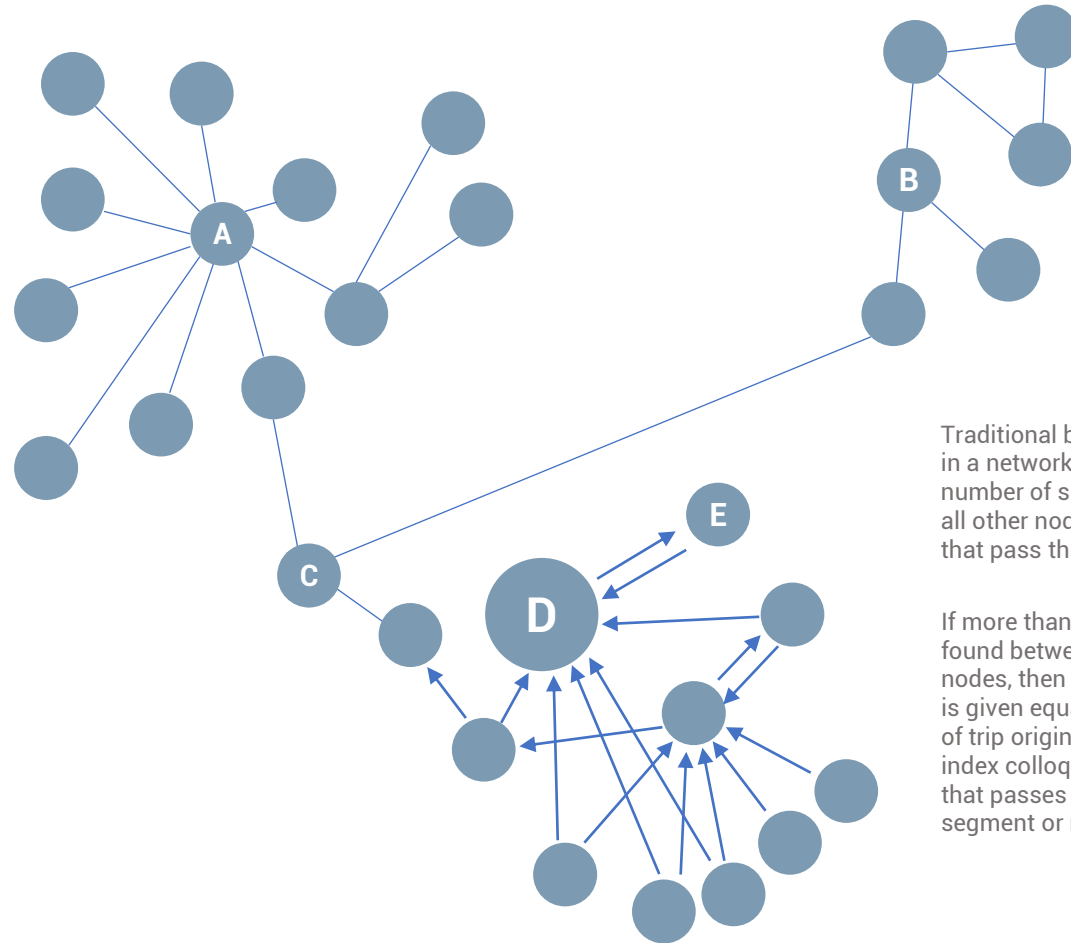
Which node can easily reach all other nodes in graph or subgraph?

"B" is closest with the fewest hops in its subgraph

"D" is foremost based on number and weighting of in-link

Betweenness

Which node has the most control overflow between nodes and groups?
"C" is the bridge

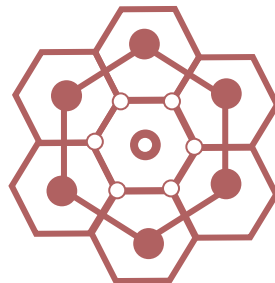


Traditional betweenness of a node in a network is defined as the number of shortest paths between all other node pairs in the network that pass through that node

If more than one shortest path is found between two surrounding nodes, then each equidistant path is given equal weight. Given a set of trip origins and destinations, the index colloquially captures the flow that passes through each network segment or node.

These models rely on the assumption that the environment is a featureless plain, where people can approach building along straight-line paths from any point in the area. **Its a simplification that clarifies the analysis and produces a more parsimonious model.** The reality of built environments is more complex—a person walking to a store/office from residents is forced to follow a network of streets, take turns when routes intersect, and choose a path among multiple competing alternatives.

Christaller

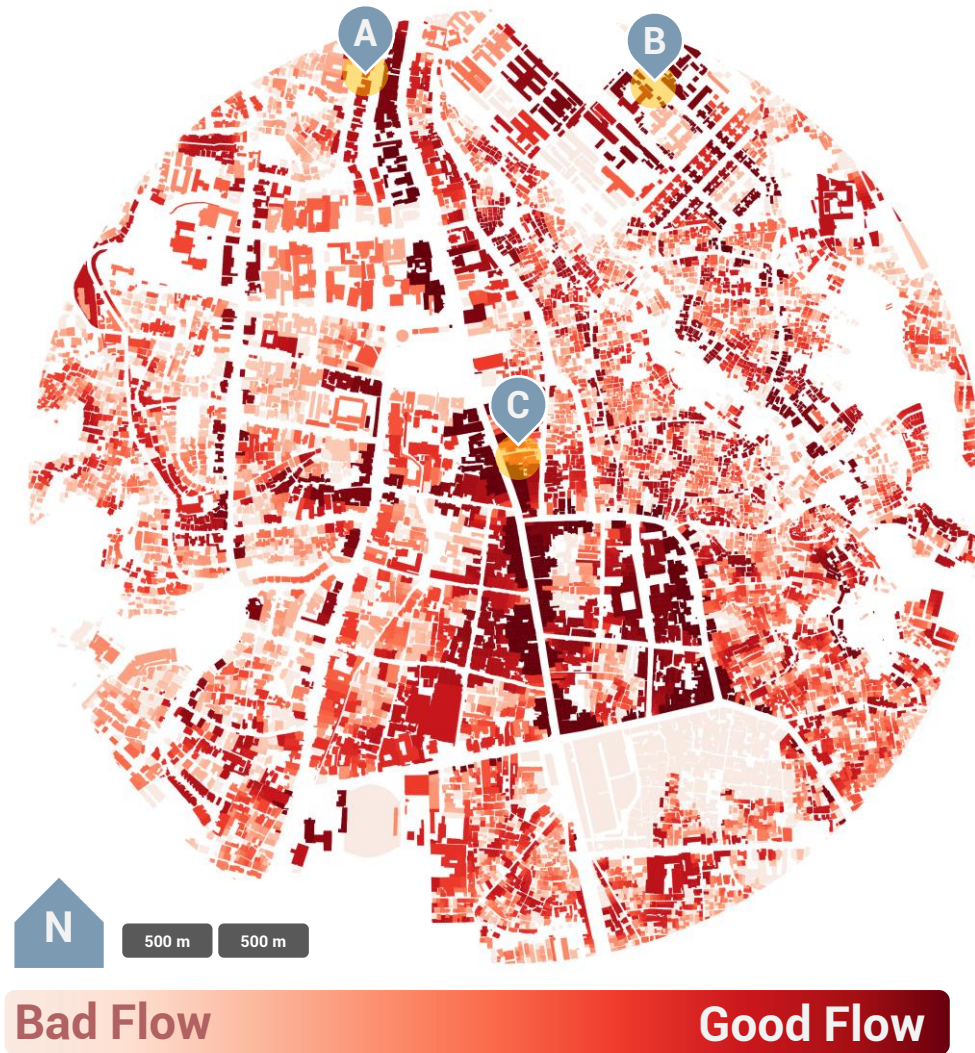


Alonso's bid-rent model: the value of the central location in a retail/office building does not arise from routine centrifugal trips from the periphery to the center, but from foot traffic that leads from one peripheral anchor to another. In other words the value of betweenness in retail building lies in the fact that these locations are between other stores and can be visited unplanned, with little extra effort. In simple spatial structures of store building, the betweenness and centrality of a location often coincide. However, in a more complex network of city streets, the two qualities of location can and do significantly diverge.

Alonso

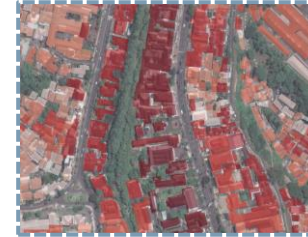


The Result



Tanggul Taman Kota

A



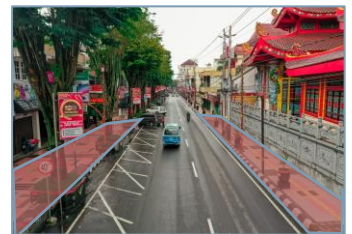
Rindam Area

B



Pecinan Area

C



Key Factors Analysis

Access to residents was relatively unimportant and access to store varied widely, but virtually all store categories were found locating at places of intense "betweenness" and passing traffic.



More complex network of city streets, more barrier of pedestrian flow. Even with buildings in close proximity, a high density of intersections in an area indicates a notable decrease in pedestrian flow, hindering connections between buildings.



Space of Building Block

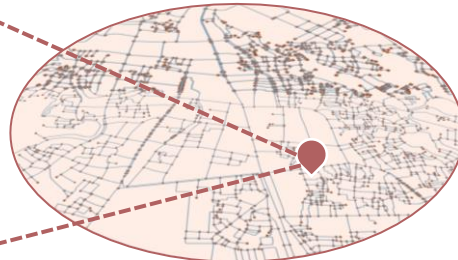
Spatial relationships in dense urban settings are complex and diverse, with no simple solutions for incentivizing desired land use patterns. But exerting careful controls over multiple simultaneous factors can indeed give planners leverage to improve the pedestrian environment in effective ways.



Commerce & Services Zone



Settlement Zone



The analysis indicates that the influence of clustering characteristics in the commercial and service areas in Magelang City serves as a catalyst in enhancing pedestrian flow. Across all structures encompassing commerce and services, clustering significantly impacts the strategic patterns of pedestrian movement, irrespective of the location-specific characteristics in that area.

